AIDS-2 IA 1 QB

Q1 (2 Marks Each)

1. Perform the operations of union, intersection, difference, and complement on given fuzzy sets.
2. Compute the algebraic sum, algebraic product, bounded sum, and bounded difference for given fuzzy sets.
3. List and explain the various fuzzification methods used for assigning membership values.
4. List and explain the different defuzzification methods.
5. Explain the `Markov decision process` in detail.
6. Explain `Bayes' theorem` and its significance.
7. Explain the centroid method of defuzzification.
8. Given a joint distribution table, perform inference to compute the required probability.
9. Explain the Bayesian belief network with a suitable example, such as the Harry Burglar Alarm system.
10. List and explain the advantages and disadvantages of a cognitive system.
11. A single card is drawn from a standard deck of `52` playing cards. Given that the probability of drawing a `king` is `4/52`, calculate the posterior probability `P(King | Face)`, which is the probability that the drawn card is a king given that it is a face card.
12. Differentiate between fuzziness and probability.

Q2 (5 Marks Each)

1. Explain the elements that constitute a cognitive system.
2. Describe the steps involved in building typical cognitive applications, and explain the same for a healthcare application.
3. With the help of a neat diagram, explain the various phases involved in natural language processing.

4. Discuss the applications of fuzzy logic in different domains.
5. Discuss the foundation of cognitive computing and highlight its distinguishing features.
6. Discuss the limitations of the deterministic approach and explain how probabilistic reasoning overcomes these limitations with a suitable example.
7. Describe the role of utility and expected utility in decision theory, and illustrate with an example.

Q3 (5 Marks Each)

1. Design a fuzzy logic controller to determine the wash time of a domestic washing machine.
2. Using the Mamdani fuzzy model, design a fuzzy logic controller to regulate the temperature of a domestic shower (up to step 3).
3. Design a fuzzy logic controller for a water purifier (up to step 3).
4. Given two fuzzy relations, calculate the max-min composition.
5. Given two fuzzy relations, calculate the max-product composition.